US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250277632
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	Patel; Brijeshkumar et al.

SEALS AND FLOW RESTRICTORS FOR ROTARY MACHINES

Abstract

A rotary machine includes a rotor with a plurality of plates defining openings therebetween, a housing enclosing the rotor, and a flow restrictor. The flow restrictor includes a body attached to a plate of the plurality of plates of the rotor and extending from the rotor toward the housing. The flow restrictor also includes a plurality of extensions extending from the body toward the housing to form a groove between extensions of the plurality of extensions.

Inventors: **Patel; Brijeshkumar (Glasgow, GB), Mochar; Ryan (Cumbernauld, GB), Whyte; William Murray (Glasgow, GB), Moore; Joanne (Falkirk, GB), Hogg; Dougal (Glasgow, GB)**

Applicant: **Howden Group Limited** (Scotland, GB)

Family ID: **95022795**

Appl. No.: **19/063834**

Filed: **February 26, 2025**

Related U.S. Application Data

us-provisional-application US 63560127 20240301

Publication Classification

Int. Cl.: **F28D19/04** (20060101)

U.S. Cl.:

CPC **F28D19/047** (20130101); F28F2230/00 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/560,127, filed Mar. 1, 2024, which is hereby incorporated by reference in its entirety for all purposes.

FIELD OF INVENTION

[0002] The present invention relates to sealing and flow restrictor arrangements between zones of rotary heat exchangers, rotary adsorption machines and the like.

BACKGROUND

[0003] Rotary adsorption machines (RAMs), which are also known as thermal swing adsorption machines, pressure swing adsorption machines, regenerative rotary separators, and the like, are often deployed to recover specific gasses, elements, and/or particulates, such as carbon dioxide. More specifically, RAMs are often deployed for point source carbon capture and/or for direct air carbon capture. In any case, RAMs typically include an adsorbent material, such as activated carbon, metal-organic frameworks (MOFs) or zeolite (e.g., hydrated aluminosilicates of alkaline and alkaline-earth metals), in a rotatable rotor.

[0004] Some RAMs include a cylindrical section that is configured to circumferentially surround the rotor and to define a plurality of zones through which the rotor can rotate. A plurality of ducts may define passageways into and out of the plurality of zones, and the cylindrical section may connect an inlet of each duct of the plurality of ducts to an outlet of each duct of the plurality of ducts. These zones can include an adsorption zone, a desorption zone, and a regeneration zone that typically operate at different temperatures and pressures. Two or more sector plate assemblies may define and/or separate adjacent zones. Meanwhile, in the rotor, radial plates extend between a central hub and an outer shell of the cylindrical section to at least partially define containers within which the adsorbent material is retained.

[0005] During operation of the RAM, a process gas, such as a carbon dioxide (CO.sub.2)-laden gas, enters the rotor, and the target gasses, elements, and/or particulates (e.g., CO.sub.2) is/are adsorbed onto the adsorbent material. The rotor then rotates the adsorbed substance into a desorption zone to release the target substance from the adsorbent so that the target substance can be captured, processed, or used. The desorption is caused by a change in pressure and/or a change in temperature (e.g., by passing steam through the rotor and/or through electric heating elements). Since these different zones are capturing and releasing a target substance and operate at different pressures, it is quite important to provide proper seals or flow restrictors between the zones to limit or eliminate fluid flow/leakage between them.

SUMMARY

[0006] Disclosed are apparatuses for minimizing or eliminating leakage between adjacent zones of a rotary machine, such as a RAM. That is, while the present disclosure primarily describes seal applications for RAMs, the sealing arrangements disclosed herein are equally applicable to minimizing or eliminating leakage between adjacent zones of a rotary heat exchanger and like machines.

[0007] According to some implementations, a rotary machine is provided. The rotary machine includes a rotor with a plurality of plates defining openings therebetween, a housing enclosing the rotor, and a flow restrictor. The flow restrictor includes a body attached to a plate of the plurality of plates of the rotor and extending from the rotor toward the housing. The flow restrictor also includes a plurality of extensions extending from the body toward the housing to form a groove between extensions of the plurality of extensions.

[0008] According to some implementations, a flow restrictor of a rotary machine is provided. The flow restrictor includes a body configured to attach to a rotor of the rotary machine. The rotor

includes a plurality of plates, and the body being configured to attach to a plate of the plurality of plates. The flow restrictor also includes a plurality of extensions extending from the body toward a housing of the rotary machine to inhibit fluid flow between the housing and the plate.

[0009] According to some implementations, a flow restrictor of a rotary machine is provided. The flow restrictor includes a body configured to attach to opposite sides of a plate of a rotor of the rotary machine and an extension extending from a distal end of the body, wherein the extension is configured to flex toward a housing of the rotary machine in response to a pressure differential across the rotor to provide a seal against the housing.

[0010] According to some implementations, a RAM is provided that comprises an adsorption zone, a desorption zone, and a regeneration zone. A sector plate having a first surface is disposed between adjacent or adjoining zones. The RAM includes a rotatable rotor having a plurality of spaced-apart radial plates that may at least partially define containers within which an adsorbent material may be retained. Each of the plurality of radial plates has a second surface that is configured to intermittently face towards the first surface of the sector plate as the rotor rotates. Each of the radial plates has a first side and a second side opposite the first side. It is appreciated that the sealing and flow restrictor apparatuses disclosed herein are applicable for use between other types of zones found in RAMs, and are also applicable for use in other equipment comprising adjacent zones that operate at different pressures.

[0011] According to some implementations, a sealing member is coupled to the radial plate and extends over at least a portion of the first surface of the sector plate. The sealing member includes a base that is attached to or coupled to the radial plate. A body of the sealing member that extends from the base towards the sector plate includes a first flexible flap located on the first side of the radial plate. The first flexible flap is transitional between a first state and a second state. When in the first state, the first flexible flap is spaced-apart from the first surface of the sector plate. The first flexible flap is configured such that when a pressure on the first side of the radial plate is greater than a pressure on the second side of the radial plate, the first flexible flap flexes to assume the second state wherein a portion of the first flexible flap contacts the first surface of the sector plate to minimize or prevent fluid leakage across the radial plate.

[0012] According to some implementations, the body of the sealing member further includes a second flexible flap located on the second side of the radial plate which is also transitional between a first state and a second state. When in the first state, the second flexible flap is spaced-apart from the first surface of the sector plate. The second flexible flap is configured such that when the pressure on the second side of the radial plate is greater than the pressure on the first side of the radial plate, the second flexible flap flexes to assume the second state wherein a portion of the second flexible flap contacts the first surface of the sector plate to minimize or prevent fluid leakage across the radial plate. In some of these instances, the first flexible flap and/or the second flexible flap changes shape when transitioning between the first and second states. Additionally or alternatively, when in the first state, the first flexible flap and/or second flexible flap has a convex surface and a concave surface opposite the convex surface with the convex surface facing the first surface of the sector plate. According to one such implementation, the first flexible flap and/or second flexible flap are/is wing-shaped at least when in the first state.

[0013] According to alternative implementations, fluid flow restriction across the radial plate is achieved through the use of a flow restrictor that is attached or coupled to the radial plate. The flow restrictor has a body that resides between the second surface of the radial plate and the first surface of the sector plate. The flow restrictor is configured to restrict or prevent fluid flow between the adjacent zones when the second surface of the radial plate is aligned with the first surface of the sector plate. An end portion of the flow restrictor that faces the first surface of the sector plate includes a plurality of walls that are spaced apart from one another to form a plurality of spaced-apart grooves that are arranged side-by-side in a width direction of the flow restrictor and a width direction of the sector plate. Each of the plurality of grooves has an opening that faces the first surface of the sector plate. Preferably, no portion of the flow restrictor contacts the sector plate such that a gap continuously exists between the distal end of the flow restrictor and the sector plate. This latter feature advantageously eliminates a wearing of the flow restrictor through contact with the sector plate.

[0014] According to some implementations, the plurality of walls includes first and second angled outermost walls with the first angled outermost wall extending outward in a direction of the first side of the radial plate and towards the first surface of the sector plate, and with the second angled outermost wall extending outward in a direction of the second side of the radial plate and towards the first surface of the sector plate. According to some implementations, the flow restrictor is configured such that when a pressure on the first side of the radial plate is greater than a pressure on the second side of the radial plate, a flow separation bubble is formed on a surface of the first angled outermost wall that faces the first surface of the sector plate. According to some implementations, the flow restrictor is also configured such that when a pressure on the second side of the radial plate is greater than a pressure on the first side of the radial plate, a flow separation bubble is formed on a surface of the second angled outermost wall that faces the first surface of the sector plate. The flow separation bubbles advantageously minimize or impede fluid flow across the width of the flow restrictor.

[0015] According to some implementations, the flow restrictor is configured such that fluid flow expansion losses occur at the entrances of one or more of the plurality of grooves, and fluid flow contraction losses occur between one or more sets of grooves when a pressure on the first side of the radial plate is greater than a pressure on the second side of the radial plate, or when a pressure on the second side of the radial plate is greater than the pressure on the first side of the radial plate. The direction of fluid flow is always from the high-pressure side to the low-pressure side. Therefore, when the pressure difference alternates, the position of the entrances at the grooves alternates correspondingly.

[0016] According to some implementations, the flow restrictor is configured such that fluid flow vortices form inside one or more of the plurality of grooves when a pressure on the first side of the radial plate is greater than a pressure on the second side of the radial plate, or when a pressure on the second side of the radial plate is greater than the pressure on the first side of the radial plate. The formation of flow vortices inside one or more of the grooves advantageously minimizes or impedes fluid flow across the width of the flow restrictor.

[0017] According to other implementations, the minimizing or prevention of fluid flow between the adjacent zones is achieved through the use of a sealing member that includes a metallic spring assembly having first and second undulating side sections that are arranged symmetrical to one another. End portions of the undulating side sections are attached or coupled to the radial plate. A second end portion of the metallic spring assembly is configured to press against the first surface of the sector plate when the second surface of the radial plate is aligned with the first surface of the sector plate to minimize or prevent fluid flow between the adjacent zones. According to some implementations, the second end portion includes leading and trailing ramp surfaces that facilitate a smooth compression and decompression of the spring assembly as the sealing member respectively approaches and moves away from the sector plate. When the second surface of the radial plate is aligned with the first surface of the sector plate, at least a portion of the leading ramp surface is located on the first side of the radial plate and at least a portion of the trailing ramp surface is located on the second side of the radial plate. Importantly, the symmetric arrangement of the sealing member continuously urges the second portion of the sealing member towards a vertical alignment with the radial plate. This advantageously urges the second portion of the sealing member towards a flush relationship with the first surface of the sector plate.

[0018] According to other implementations, the minimizing or prevention of fluid flow between the adjacent zones is achieved through the use of a sealing member that includes a metallic spring assembly and a non-spring element. The metallic spring assembly includes first and second

undulating side sections that are arranged symmetrical to one another. End portions of the undulating side sections are attached or coupled to the radial plate. A proximal end portion of the non-spring element is coupled to and supported by the metallic spring assembly. A distal end portion of the non-spring element has a surface that is configured to press against the first surface of the sector plate when the second surface of the radial plate is aligned with the first surface of the sector plate. The non-spring element includes leading and trailing ramp surfaces that facilitate a smooth compression and decompression of the metallic spring assembly as the sealing member respectively approaches and moves away from the sector plate. When the second surface of the radial plate is aligned with the first surface of the sector plate at least a portion of the leading ramp surface is located on the first side of the radial plate and at least a portion of the trailing ramp surface is located on the second side of the radial surface. The symmetric arrangement of the metallic spring assembly advantageously urges the non-spring element towards a vertical alignment with the radial plate.
[0019] According to some implementations, the non-spring element is a made of a rigid material that does not deform when the surface of the distal end portion of the non-spring element is pressed against the first surface of the sector plate. An advantage of using the non-spring element is that it may be made of a material that is more wear resistant then the spring assembly and can be coupled to the spring assembly in a way that makes it more readily replaceable as will be discussed in more detail below.
[0020] These and other advantages and features will become evident in view of the drawings and detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0021] To complete the description and in order to provide for a better understanding of the present invention, a set of drawings is provided. The drawings form an integral part of the description and illustrate implementations of the present invention, which should not be interpreted as restricting the scope of the invention, but just as an example of how the invention can be carried out. The drawings comprise the following figures:
[0022] FIG. **1** is a schematic view of a combined cycle power plant with a rotary adsorption machine (RAM) formed in accordance with an example implementation.
[0023] FIG. **2** is a top, front perspective view of a RAM formed in accordance with an example implementation.
[0024] FIG. **3** is a partially cut-away perspective view of a portion of the RAM of FIG. **2**.
[0025] FIG. **4**A is a cross-sectional view of an embodiment of a sealing assembly that includes an elastomeric member attached to or otherwise coupled to a radial plate of the RAM of FIG. **2**, and flaps of the elastomeric member are in a first state in which each flap is spaced apart from the sector plate.
[0026] FIG. **4**B shows the sealing assembly of FIG. **4**A with there being a pressure differential between sides of the radial plate to result in the flap on a high pressure side being flexed to press against a surface of a housing.
[0027] FIG. **5**A is a cross-sectional view of an embodiment of a flow restrictor located between a radial plate and a housing in which one or more of fluid separation, fluid expansion loss, or fluid contraction loss is induced to minimize or prevent fluid flow leakage between adjacent zones of a RAM.
[0028] FIG. **5**B illustrates the flow restrictor of FIG. **5**A, showing areas of fluid separation, fluid flow expansions, fluid flow contractions, and fluid vortices induced by the configuration of the flow restrictor.

[0029] FIG. **6** is a cross-sectional view of another embodiment of a flow restrictor that is configured to induce one or more of fluid separation, fluid expansion loss, or fluid contraction loss to minimize leakage between adjacent zones of a RAM.
[0030] FIG. **7** is a cross-sectional view of yet another embodiment of a flow restrictor that is configured to induce one or more of fluid separation, fluid expansion loss, or fluid contraction loss to minimize leakage between adjacent zones of a RAM.
[0031] FIG. **8** is a cross-sectional view of still another embodiment of a flow restrictor that is configured to induce one or more of fluid separation, fluid expansion loss, or fluid contraction loss to minimize leakage between adjacent zones of a RAM.
[0032] FIG. **9** is a cross-sectional view of an embodiment of a symmetrical spring structure arranged between a radial plate and a housing of a RAM. The spring structure includes leading and trailing ramp sections and grooves located between the ramp sections.
[0033] FIG. **10** is a cross-sectional view of another embodiment of a sealing member similar to that of FIG. **9**, in which the leading ramp, trailing ramp, and grooves are not comprised in the spring structure, but in a non-spring element coupled to the spring structure.
[0034] FIG. **11** illustrates a portion of the RAM of FIG. **2** in greater detail to provide another location for placement of a sealing member.
[0035] FIG. **12** illustrates another portion of the RAM of FIG. **2** in greater detail to provide yet another location for placement of a sealing member.

DETAILED DESCRIPTION

[0036] Generally, this application is directed to a rotary adsorption machine (RAM). However, it is important to note that the sealing and flow restrictor arrangements disclosed herein are also applicable for use in rotary heat exchangers, including low temperature rotary heat exchangers (e.g., those operating at less than 200° C.) and cold temperature rotary heat exchangers. Disclosed herein are sealing arrangements to reduce or eliminate the leakage of fluids (gases and/or liquids) between the zones of a RAM, a rotary heat exchanger, and the like for the purpose of increasing their effectiveness.
[0037] An example power plant **10** of a type that may incorporate a RAM **26** formed in accordance with the present application is illustrated in FIG. **1**. However, to be clear, the power plant **10** of FIG. **1** is merely an example and, in other implementations, RAM **26** may be positioned in any desirable location, e.g., for carbon capture. For example, power plant **10** generally depicts a combined cycle gas turbine (CCGT) power plant, but the RAM **26** could also be positioned/included in a conventional coal powered power plant or any other flue system (e.g., for point source capture). In fact, it is envisioned that the RAM **26** presented herein may be configured to capture carbon dioxide from ambient air. That is, the RAM **26** presented herein may be positioned in locations in which CO.sub.2-laden gas entering the RAM **26** is ambient air (as opposed to a process effluent).
[0038] That said, in FIG. **1**, the power plant **10** includes a gas turbine **16**, a Heat Recovery Steam Generator (HRSG) **14**, and a generator **18** coupled with a steam turbine **23**. Turbines **16** and **23** combine to drive the generator **18** to produce electricity. The steam turbine **23** is connected to a condenser **19** with an intake **20** and exhaust **22**. The power plant **10** also includes fans **24***a* and **24***b*, which may be used to move air through this system. Meanwhile, a heat exchanger **12** may be positioned adjacent the exhaust of the HRSG **14**. Although not shown, a power plant utilizing the RAM **26** might also include another heat exchanger to heat the air entering a boiler. For example, such a heat exchanger might heat air entering a boiler with heat from combustion gases expelled from the boiler (while also cooling the gas expelled from the boiler).
[0039] As shown in FIG. **1** and in combination with FIG. **2**, which illustrates the RAM **26** of FIG. **1** in further detail, the cooled exhaust gas enters the RAM **26** as a first flow F**1** and enters the RAM **26** via a first duct **110**. However, to reiterate, exhaust gas is merely one example of gas that may enter the RAM **26** as first flow F**1**. As other examples, the first flow F**1** may be a flow of ambient

air and/atmosphere, or a combination of ambient air/atmosphere and an exhaust gas. In any case, when the first flow F**1** encounters a rotor **34** included in the RAM **26**, adsorptive elements in the rotor **34** can adsorb a specific portion of the first flow F**1** (e.g., carbon dioxide). Then, the adsorptive elements in the rotor **34** can carry the adsorbed portion of the first flow F**1** through a partial rotation. Meanwhile, a portion of the first flow F**1** that is not captured by the adsorptive elements may exit the RAM **26** as process flow F**1**′, e.g., to (or back to) atmosphere, e.g., by way of heat exchanger **12** where it can be used to cool exhaust gas. Additionally or alternatively, the process flow F**1**′ could be fed to a conduit that directs the process flow F**1**′ to a downstream processing operation that requires clean gas/air. The area of the rotor **34** aligned with the first flow F**1** may generally be referred to herein as a first zone Z**1** (i.e., an adsorptive zone Z**1**) of the RAM **26**.

[0040] As the rotor **34** rotates, it moves the adsorbed portion of the first flow F**1** (e.g., carbon dioxide) out of the adsorptive zone Z**1** (e.g., by rotating the adsorptive elements that have adsorbed the portion of the first flow F**1**) and into a second zone Z**2** (i.e., a desorption zone Z**2**) of the RAM **26**. In the desorption zone Z**2**, a second flow F**2** is directed into the RAM **26** to cause the adsorptive elements of rotor **34** carrying the adsorbed portion of the first flow F**1** to desorb the adsorbed portion of the first flow F**1**. For example, steam may be directed into the RAM **26** as the second flow F**2** to create a temperature change that releases carbon dioxide from adsorptive elements for carbon capture. To illustrate this example, the steam of the second flow F**2** emanates from steam turbine operations (e.g., from condenser **19**) in FIG. **1**. In any case, the adsorbed portion of the first flow F**1** is released into the second flow F**2** to generate a flow F**2**′ exiting RAM **26**. By way of example, the flow F**2**′ may carry carbon dioxide and may be directed to a storage tank, condenser, and/or stripper, e.g., to prevent the carbon dioxide from entering or re-entering the atmosphere (e.g., to remove carbon dioxide from the atmosphere).

[0041] After adsorptive elements desorb the adsorbed portion of the first flow F**1** (e.g., carbon dioxide), the adsorptive elements may move into a third zone Z**3** (i.e., a regeneration zone Z**3**). In the third zone Z**3**, conditioning air (e.g., driven by fan **24***a*) may flow through the RAM **26** to “regenerate” the adsorptive elements, entering as flow F**3** and exiting as flow F**3**′(which, may, in some instances, combine with the process flow F**1**′ on exiting the RAM **26**, as shown in FIG. **1**). This conditioning air prepares the adsorptive elements to re-enter the adsorption zone Z**1** (e.g., by cooling the adsorptive elements) so that the adsorptive elements can continue cycling through the three zones of the RAM **26**. That is, continued rotation of a particular adsorptive element of rotor **34** through a full 360° rotation within the RAM **26** will cause the particular adsorptive element to adsorb a specific portion of the first flow F**1**, desorb the flow component, and regenerate. Thus, a cylindrical rotor **34** full of adsorptive elements will continuously capture a component/portion of a first flow of gas F**1** entering the RAM **26**.

[0042] However, to be clear, the RAM **26** illustrated in the figures of this application is merely an example and other implementations may include any number of variations. For example, a RAM **26** formed in accordance with the present application may include any number of zones, e.g., to incorporate isolation zones, multiple stages or regeneration, desorption, and/or adsorption, or for any other reason. Additionally or alternatively, the various flows entering and exiting the RAM **26** may emanate from any desirable source or flow to any desirable location, including a source of that flow or another flow (e.g., to recycle flows of fluid). As yet another example, the composition of the various flows can be varied, such as by using a fluid flow other than steam for desorption.

[0043] FIG. **3** illustrates the RAM **26** of FIG. **2** in greater detail by providing a cut-away view of a portion of the RAM **26**. FIGS. **2** and **3** are discussed together to describe the RAM **26**. At a high-level, the RAM **26** includes a rotor **34** that is rotatable within a housing **100**. The housing **100** is specifically designed to enclose and seal against portions of the rotor **34** to help dictate how and where fluid (e.g., gas) will enter, exit, or move with the rotor **34**. As mentioned, the RAM **26** presented herein, including the housing **100** and rotor **34**, may be particularly suitable for large scale (e.g., industrial) operations. Thus, in at least some instances, the rotor **34** may have a diameter equal to or greater than 20 meters, such as 24 meters, and the housing **100** may be sized accordingly.

[0044] As can be seen in FIG. **3**, the rotor **34** includes a central hub **36** and an outer shell **35**. Radial plates **37** extend between the central hub **36** and the outer shell **35** and are offset from one another to at least partially define containers or openings **40** therebetween. The containers **40** are configured to receive and retain adsorbent material. In at least some implementations, the rotor **34** also includes circumferential plates to subdivide the containers **40**. Either way, adsorbent material may be stored and/or installed within the containers **40**. For example, adsorbent material may be "dropped down" into the containers **40** to fill the rotor **34** with adsorbent material. In at least some instances, the adsorbent may be formed from any adsorbent now known or developed hereafter that is suitable for adsorbing carbon dioxide, such as activated carbon, MOFs, zeolite(s), or combinations thereof.

[0045] As mentioned, the rotor **34** is configured to continuously rotate around the central hub **36** to move radially aligned containers **40** through a cycle of zones (e.g., through zones Z**1**, Z**2**, and Z**3**). During this rotation, the housing **100** is generally designed to circumferentially retain gas in the rotor **34** and to create pathways along which fluid can axially enter or exit the rotor **34**. The circumferential retention can be achieved by closely positioning a cylindrical section **108** of the housing **100** against the outer shell **35** of the rotor **34**. Additionally, sector plates located between the zones of the RAM are equipped with features that facilitate producing a seal between the zones for the purpose of minimizing or eliminating fluid flow between the zones. FIG. **2** shows an example sector plate **29** positioned between the adsorption zone (Z**1**) and the regeneration zone (Z**3**) and above the radial plates **37**. In the depicted implementation, a first sector plate **29** separates the adsorption zone Z**1** (generally aligned with first duct **110**) from at least the regeneration zone Z**3**.

[0046] The examples disclosed below are directed to sealing arrangements between the first sector plate **29** and radial plates **37**. Moreover, the examples are directed to sealing arrangements between an end surface **37***a* (e.g., an outer surface, a top surface, see, e.g., FIG. **4**A) of the radial plates **37** with a first surface **29***a* (e.g., an inner surface, a bottom surface, a lower surface, see, e.g., FIG. **4**A) of the first sector plate **29**. A second sector plate (not shown in the figures), similar to and horizontally aligned with the first sector plate **29** is typically disposed below the radial plates **37**. The sealing arrangements disclosed herein are equally applicable to producing a seal between a bottom end of the radial plates **37** with a top surface of the second sector plate. Similarly, the sealing arrangements disclosed herein may be equally applicable to longitudinally extending surfaces extending between the bottom and top sector plates (e.g., vertically extending portions of a sector assembly). With reference to FIGS. **2** and **3**, the aforementioned first and second sector plates may be respectively attached or coupled to top and bottom frame assemblies **300** and **400** of RAM **26**. Although not shown in FIG. **2**, similar sets of sector plates may be disposed between the adsorption zone (Z**1**) and desorption zone (Z**2**) and also between the desorption zone (Z**2**) and the regeneration zone (Z**3**).

[0047] With continued reference to FIGS. **2** and **3**, overall, the housing **100** extends from a front **101** to a back **102**, from a first side **103** to a second side **104**, and from a bottom **106** to a top **105**. In the depicted implementation, different streams of fluid enter or exit the RAM **26** in a generally vertical or longitudinal manner (i.e., from the bottom **106** to the top **105**, or vice versa). Thus, the housing **100**: (a) includes a cylindrical section **108** that circumferentially surrounds the rotor **34**; and (b) defines a plurality of ducts at the top **105** and bottom **106** of the RAM **26**. Specifically, in the depicted implementation, the RAM **26** includes three ducts that are generally aligned with zones Z**1**, Z**2**, and Z**3**: (1) a first duct **110** generally aligned with adsorption zone Z**1**; (2) a second duct **130** generally aligned with desorption zone Z**2**; and (3) a third duct **150** generally aligned with regeneration zone Z**3**. However, other implementations may include any number of ducts and do not necessarily need to include the same number of ducts and zones.

[0048] In the depicted implementation, the first duct **110** extends from an inlet disposed adjacent the top **105** of the housing **100** to an outlet disposed adjacent the bottom **106** of the housing **100**. Meanwhile, the second duct **130** and third duct **150** extend from inlets that are positioned adjacent the bottom **106** of the housing **100** to outlets that are respectively positioned adjacent the top **105** of the housing **100**. Thus, the first flow F**1** entering the first duct **110** generally flows in a first longitudinal direction (e.g., downwards) while flows F**2** and F**3** entering ducts **130** and **150** generally flow in an opposite longitudinal direction (e.g., upwards). As specific examples, the first flow F**1** may comprise ambient air and/or a process effluent flowing downwards into the rotor **34** via the inlet of the first duct **110** while the second flow F**2** comprises steam flowing upwards into the rotor **34** via the inlet of second duct **130** and the third flow F**3** comprises conditioning air flowing upwards into rotor **34** via the inlet of third duct **150**.

[0049] In the examples that follow, reference is made to sealing members and flow restrictors located between the end surfaces **37***a* of the radial plates **37** and the first surface **29***a* of the sector plate **29** situated between the adsorption zone (Z**1**) and the regeneration zone (Z**3**). It is appreciated that such sealing arrangements are equally applicable to radial plate/housing interfaces in other locations of the RAM **26**, including locations that prevent or inhibit leakage between adjacent zones and/or circumferential leakage around a rotor and/or matrix. That is, the sealing members and flow restrictors may be coupled or attached to any of the upper and lower ends of the radial plates **37** to seal against one or more other plates located in the RAM **26** but may also be coupled or attached to circumferential sections, axial plates, axial ends of radial plates, etc.

[0050] FIGS. **4**A and **4**B are each a cross-sectional view of a sealing arrangement **500** according to one implementation in which the sector plate **29** is aligned with one of the plurality of radial plates **37** via rotation of a rotor of a RAM (e.g., the RAM **26**). That is, an end surface **37***a* (also referred to herein as "second surface") of the radial plate **37** is situated facing the first surface **29***a* of the sector plate **29**. The radial plate **37** has first and second opposite sides **37***b* and **37***c* that respectively reside in zones Z**1** and Z**3** of the RAM. For discussion purposes, zone Z**1** will be assumed to operate at a higher pressure than zone Z**3**. As previously discussed, the radial plate **37** is one of a plurality of spaced-apart radial plates that rotates inside a housing of the RAM. Arrow "R" indicates the direction of rotation.

[0051] The sealing arrangement **500** includes a sealing member **501** attached to the radial plate **37**. The sealing member **501** includes a base **502** that is attached to or otherwise coupled to the radial plate **37** by any suitable attachment means which may include the use of bolts, screws, an adhesive, an interference fit, etc. By way of example, the base **502** may capture the sides **37***b* and **37***c* of the radial plate **37**.

[0052] The body **504** of the sealing member **501** extends from the base **502** towards the sector plate **29** and includes a first flexible flap or extension **510** (e.g., a leading flexible flap) located primarily on the first side **37***b* of the radial plate **37**. The first flexible flap **510** is transitional between a first state **530** as shown in FIG. **4**A and a second state **532** as shown in FIG. **4**B. When in the first state **530**, the first flexible flap **510** is spaced-apart from the first surface **29***a* of the sector plate **29** by a distance d**1**, which may be in the range of 0 millimeters and 4 millimeters. The first flexible flap **510** is configured such that a pressure differential in which the pressure on the first side **37***b* of the radial plate **37** is greater than the pressure on the second side **37***c* of the radial plate causes the first flexible flap **510** to flex and assume the second state **532** wherein a portion of the first flexible flap **510** contacts the first surface **29***a* of the sector plate **29** to minimize or prevent fluid leakage between zones Z**1** and Z**3**.

[0053] According to some implementations, the body **504** of the sealing member **501** further includes a second flexible flap or extension **520** (e.g., a trailing flexible flap) located primarily on the second side **37***c* of the radial plate **37**. When in the first state, the second flexible flap **520** is spaced-apart from the first surface **29***a* of the sector plate **29**. The second flexible flap **520** is configured such that a pressure differential in which the pressure on the second side **37***c* of the radial plate **37** is greater than the pressure on the first side **37***b* of the radial plate **37** causes the second flexible flap **520** to flex and assume another second state **532** wherein a portion of the second flexible flap contacts the first surface **29***a* of the sector plate **29** to minimize or prevent fluid leakage between zones Z**1** and Z**3**.

[0054] As illustrated in FIGS. **4**A and **4**B, the first flexible flap **510** has a first shape when in the first state **530** and a second shape different than the first shape when in the second state **532**. In implementations wherein the sealing member **501** includes the second flexible flap **520**, the second flexible flap **520** may additionally or alternatively change shape when transitioning between its first and second states **530**, **532**.

[0055] As shown in FIG. **4**A, according to some implementations, the first and second flexible flaps **510**, **520** have a same or similar shape when in their first states **530**. Although not shown in the figures, according to some implementations, the first and second flexible flaps **510**, **520** have a same or similar shape when in their second states.

[0056] According to some implementations, the first and second flexible flaps **510**, **520** are wing-shaped when in their first state **530** as shown in FIG. **4**A and may or may not be wing-shaped when in their second state **532**. According to some implementations, the first flexible flap **510** has a convex surface **511** and a concave surface **512** opposite the convex surface **511**, with the convex surface **511** facing the first surface **29***a* of the sector plate **29**. In implementations wherein the sealing member **501** includes the second flexible flap **520**, the second flexible flap **520** has a convex surface **521** and a concave surface **522** opposite the convex surface **521** with the convex surface **521** facing the first surface **29***a* of the sector plate **29**. The convex surface **511**, **521** of each of the first and second flexible flaps **510**, **520** is preferably curved to facilitate a smooth engagement and disengagement of the flexible flaps **510**, **520** with the sector plate **29**. For the same reason, according to some implementations, at least the first flexible flap **510** includes a leading curved surface **514**. Additionally, the curved profile of the flexible flaps **510**, **520** forms a groove **524** therebetween. In some embodiments, the groove **524** can further help restrict fluid leakage between zones Z**1** and Z**3**. By way of example, the groove **524** may provide a discontinuity that disrupts fluid flow between the sector plate **29** and the sealing member **501** along the flexible flaps **510**, **520**.

[0057] To minimize wear of the sector plate **29** and the sealing member **501**, at least portions of the first and second flexible flaps **510**, **520** that are intended to contact the sector plate may possess a lubricious coating. This coating may facilitate movement of the flexible flaps **510**, **520** along the sector plate **29** and reduce potential abrasion of the sector plate **29** against the flexible flaps **510**, **520**. The lubricious coating may comprise, for example, polytetrafluoroethylene (PTFE).

[0058] According to some implementations, each of the first and second flexible flaps **510**, **520** is made of an elastomeric material that enables it to automatically transition from its second state **532** towards its first state **530**. That is, the elastomeric material is able to bend or deform when a load (e.g., a pressure) is applied to it, and the elastomeric material is able to fully recover or substantially recover its original shape when the load is removed. According to other implementations, the first and second flexible flaps **510**, **520** are respectively made of first and second flexible strips of spring metal to be able to bend in response to a load and recover its original shape absent the load. According to some implementations, the sealing member **501** is a unitary structure (i.e., is made from a single piece of material).

[0059] FIG. **5**A illustrates a cross-sectional view of a flow restrictor **600** located between the radial plate **37** and the sector plate **29** in which one or more of fluid separation, fluid expansion loss, or fluid contraction loss is induced to minimize or prevent fluid flow leakage between adjacent zones of a RAM. FIG. **5**B illustrates the flow restrictor of FIG. **5**A, showing areas of fluid separation, fluid flow expansion, fluid flow contraction, and fluid vortices induced by the configuration of the flow restrictor.

[0060] According to some implementations, fluid flow restriction across the radial plate **37** is achieved through the use of the flow restrictor **600** that is attached or coupled to the radial plate **37**. The flow restrictor **600** has a body **601** that resides between the second surface **37***a* of the radial plate **37** and the first surface **29***a* of the sector plate **29**. For example, the body **601** may be attached to the radial plate **37** by capturing the sides **37***b* and **37***c* of the radial plate **37** and/or via the use of bolts, screws, an adhesive, an interference fit, etc. The flow restrictor **600** is configured to restrict or prevent fluid flow between the adjacent zones (e.g., zones Z**1** and Z**3**) when the second surface **37***a* of the radial plate **37** is aligned with the first surface **29***a* of the sector plate **29**. An end portion **610** of the flow restrictor **600** that faces the first surface **29***a* of the sector plate **29** includes a plurality of walls or extensions **610***a-d* that are spaced apart from one another to form a plurality of spaced-apart grooves **650***a-c* that are arranged side-by-side in a width direction "w" of the flow restrictor **600** and a width direction "w" of the sector plate **29**. Each of the plurality of grooves **650***a-c* has an opening **628** that faces the first surface **29***a* of the sector plate **29**. Preferably, no portion of the flow restrictor **600** contacts the sector plate **29** such that a gap **630** continuously exists between the distal-most end of the flow restrictor **600** (an end of the flow restrictor **600**, such as of one of the walls **610***a-d*, nearest the sector plate **29**) and the first surface **29***a* of the sector plate **29**. This latter feature advantageously eliminates a wearing of the flow restrictor **600** and the sector plate **29** that would otherwise occur through contact therebetween. According to some implementations, a distance d**2** between the distal-most end of the flow restrictor **600** and the first surface **29***a* of the sector plate **29** is in the range of 1 to 2 millimeters.
[0061] According to some implementations, the plurality of walls **610***a*-**610***d* includes first and second angled outermost walls **610***a* and **610***d*, with the first angled outermost wall **610***a* extending outward in an oblique direction of the first side **37***b* of the radial plate **37** towards the first surface **29***a* of the sector plate **29**, and with the second angled outermost wall **610***d* extending outward in an oblique direction of the second side **37***c* of the radial plate **37** towards the first surface **29***a* of the sector plate **29**. According to some implementations, the flow restrictor **600** is configured such that, when a pressure on the first side **37***b* of the radial plate **37** is greater than a pressure on the second side **37***c* of the radial plate **37** to urge fluid flow from the first side **37***b* toward the second side **37***c*, a flow F**4** is directed by the angled surface **660** of the first angled outermost wall **610***a* towards the first surface **29***a* of the sector plate **29** and in a direction opposite flow F**5** which is directed towards gap **630**. In this manner, flow F**4** deflects or impinges against the flow F**5** to alter the trajectory of the flow F**5**, thereby impeding the entry of flow F**5** into gap **630**. In addition, a flow separation bubble **680** may be formed on a distal surface **670** of the first angled outermost wall **610***a* that faces the first surface **29***a* of the sector plate **29**. According to some implementations, the same flow phenonium occurs on the opposite side of the radial plate **37** when a pressure on the second side **37***c* of the radial plate **37** is greater than a pressure on the first side **37***b* of the radial plate **37** to urge fluid flow from the second side **37***c* toward the first side **37***b*. In addition, a flow separation bubble may formed on a distal surface **662** of the second angled outermost wall **610***d* that faces the first surface **29***a* of the sector plate **29**. The formation of a flow separation bubble advantageously minimizes or impedes fluid flow across the angled outermost walls **610***a* or **610***d*, as the case may be. In the example of FIGS. **5**A and **5**B, walls **610***b* and **610***c* are arranged orthogonal to the first surface **29***a* of the sector plate **29**. According to other implementations, walls **610***b* and **610***c* may be arranged non-orthogonal to the first surface **29***a*.
[0062] According to some implementations, the flow restrictor **600** is configured such that fluid flow expansion losses or fluid flow contraction losses **682** occur at the opening **628** of one or more of the plurality of grooves **650***a-c* when there is a pressure differential between the first side **37***b* of the radial plate **37** and the second side **37***c* of the radial plate **37**. Moreover, according to some implementations, the flow restrictor **600** is configured such that fluid flow vortices **690** form inside one or more of the plurality of grooves **650***a-c* when there is a pressure differential between the first side **37***b* of the radial plate **37** and the second side **37***c* of the radial plate **37**. The formation of

fluid flow vortices **690** inside one or more of the grooves advantageously minimizes or impedes fluid flow across the width of the flow restrictor **600**, such as by deflecting or otherwise reducing fluid flow advancing along the first surface **29***a* of the sector plate **29**.

[0063] FIG. **6** illustrates a cross-sectional view of the flow restrictor **600** having walls **610***b* and **610***c* that are triangular and have a respective apex **690***a* and **690***b* that each points towards the first surface **29***a* of the sector plate **29** when the second surface **37***a* of the radial plate **37** is aligned with the first surface **29***a* of the sector plate **29**. Such a shape of the walls **610***b* and **610***c* can create flow, such as flow vortices, that impedes fluid flow across the width of the flow restrictor **600**.

[0064] FIG. **7** illustrates a cross-sectional view of another flow restrictor **700** that is configured to induce one or more of fluid separation, fluid expansion loss, or fluid contraction loss to minimize fluid leakage between adjacent zones of a RAM. Flow restrictor **700** functions similar to flow restrictor **600** and includes a plurality of walls **710***a-c* that includes first and second angled outermost walls **710***a* and **710***c* and a central wall **710***b* disposed between them. A first groove **750***a* is formed between the first angled outermost wall **710***a* and the central wall **710***b*, and a second groove **750***b* is formed between the second angled outermost wall **710***c* and the central wall **710***b*.

[0065] The first angled outermost wall **710***a* extends outward in an oblique direction of the first side **37***b* of the radial plate **37** towards the first surface **29***a* of the sector plate **29**, and the second angled outermost wall **710***c* extends outward in an oblique direction of the second side **37***c* of the radial plate **37** towards the first surface **29***a* of the sector plate **29**. According to some implementations, the central wall **710***b* is arranged orthogonal to the first surface **29***a* of the sector plate **29**.

[0066] According to some implementations, the flow restrictor **700** is configured such that when a pressure on the first side **37***b* of the radial plate **37** is greater than a pressure on the second side **37***c* of the radial plate **37**, similar to what is illustrated in FIG. 5B, a first flow is directed along a surface **720** of the first angled outermost wall **710***a* towards the first surface **29***a* of sector plate **29** and in a direction opposite to a second flow that flows in a direction towards gap **630**. The first flow causes a redirection of the second flow to restrict it from entering the gap **630**. In addition, a flow separation bubble (not shown) may be formed on a distal surface **722** of the first angled outermost wall **710***a* that faces the first surface **29***a* of the sector plate **29**. According to some implementations, the same flow phenonium occurs on the opposite side of the radial plate **37** when a pressure on the second side **37***c* of the radial plate **37** is greater than a pressure on the first side **37***b* of the radial plate **37**. According to some implementations, the flow restrictor **700** is also configured such that when a pressure on the second side **37***c* of the radial plate **37** is greater than a pressure on the first side **37***b* of the radial plate **37**, a flow separation bubble (not shown) is formed on a distal surface **724** of the second angled outermost wall **710***c* that faces the first surface **29** of the sector plate **29**. As explained above, the formation of flow separation bubbles advantageously minimizes or impedes fluid flow across the angled outermost walls **710***a* or **710***c*.

[0067] According to some implementations, the flow restrictor **700** is configured such that fluid flow expansion losses (not shown) occur at an opening **726** of one or more of the first and second grooves **750***a* and **750***b* and a fluid flow contraction loss (not shown) occurs between the first and second grooves **750***a* and **750***b* when there is a pressure differential between the first side **37***b* of the radial plate **37** and the second side **37***c* of the radial plate **37**. According to some implementations, the flow restrictor **700** is configured such that fluid flow vortices (not shown) form inside one or both of the first and second grooves **750***a* and **750***b* when there is a pressure differential between the first side **37***b* of the radial plate **37** and the second side **37***c* of the radial plate **37**. The formation of flow vortices inside one or both of the grooves **750***a* and **750***b* advantageously minimizes or impedes fluid flow across the width of the flow restrictor **700**, such as by deflecting or otherwise reducing fluid flow advancing along the first surface **29***a* of the sector plate **29**.

[0068] FIG. **8** illustrates a cross-sectional view of another flow restrictor **800** that is configured to induce one or more of fluid separation, fluid expansion loss, and/or fluid contraction loss to minimize leakage between adjacent zones of a RAM. Flow restrictor **800** functions similar to flow restrictors **600** and **700** and includes a plurality of walls **810***a-e* that includes first and second angled outermost walls **810***a* and **810***e* with walls **810***b-d* disposed therebetween. Formed between walls **810***a-e* is a plurality of grooves **850***a-d*. The plurality of grooves includes a first groove **850***a* positioned between the first angled outermost wall **810***a* and wall **810***b*, a second groove **850***b* positioned between walls **810***b* and **810***c*, a third groove **850***c* positioned between walls **810***c* and **810***d*, and a fourth groove **850***d* positioned between wall **810***d* and the second angled outermost wall **810***e*.

[0069] According to some implementations, each of walls **810***a* and **810***b* is angled outward in an oblique direction of the first side **37***b* of the radial plate **37** towards the first surface **29***a* of the sector plate **29**, and each of walls **810***d* and **810***e* is angled outward in an oblique direction of the second side **37***c* of the radial plate **37** towards the first surface **29***a* of the sector plate **29**. In the example of FIG. **8**, the third wall **810***c* is arranged orthogonal to the first surface **29***a* of the sector plate **29**. However, according to other implementations, each of walls **810***b-d* is arranged orthogonal to the first surface **29***a* of the sector plate **29**.

[0070] According to some implementations, flow restrictor **800** is configured such that when a pressure on the first side **37***b* of the radial plate **37** is greater than a pressure on the second side **37***c* of the radial plate **37**, a flow separation bubble (similar to that shown in FIG. **5**B) is formed on a distal surface **820** of the first angled outermost wall **810***a* that faces the first surface **29***a* of the sector plate **29**. According to some implementations, flow restrictor **800** is also configured such that when a pressure on the second side **37***c* of the radial plate **37** is greater than a pressure on the first side **37***b* of the radial plate **37**, a flow separation bubble (not shown) is formed on a distal surface **822** of the second angled outermost wall **810***e* that faces the first surface **29***a* of the sector plate **29**. The advantages associated with the formation of flow separation bubbles are discussed above.

[0071] According to some implementations, flow restrictor **800** is configured such that fluid flow expansion losses, like those discussed in conjunction with the example of FIGS. **5**A and **5**B, occur at an opening **824** of one or more of grooves **850***a-d* and fluid flow contraction losses occurs between one or more sets of adjacent grooves **850***a-d*, such as between grooves **850***a* and **850***b*, grooves **850***b* and **850***c*, and grooves **850***c* and **850***d* when there is a pressure differential between the first side **37***b* of the radial plate **37** and the second side **37***c* of the radial plate **37**. Similarly, according to some implementations, flow restrictor **800** is configured such that fluid flow vortices (not shown) form inside one or more or all of grooves **850***a-d* when there is a pressure differential between the first side **37***b* of the radial plate **37** and the second side **37***c* of the radial plate **37**. The formation of flow vortices inside one or more or all of the grooves **850***a-d* advantageously minimizes or impedes fluid flow across the width of the flow restrictor **800**, such as by deflecting or otherwise reducing fluid flow advancing along the first surface **29***a* of the sector plate **29**.

[0072] According to some implementations, any of flow restrictors **600**, **700**, or **800** is a unitary structure made from a single piece of material. According to some implementations, any of the flow restrictors **600**, **700**, or **800** are made of a polymeric material such as, for example PTFE and ethylene propylene diene monomer (EPDM). According to some implementations, the walls/extensions **610***a-d*, **710***a-c*, or **810***a-e* of any of the flow restrictors **600**, **700**, or **800** are rigid and are configured not to deform during normal operation of the RAM. This latter feature enables a more consistent production of flow separation bubbles, a more consistent production of flow expansion losses, and/or a more consistent production of flow contraction losses at the end portions of the flow restrictors **600**, **700**, or **800**.

[0073] According to other implementations, minimizing or preventing fluid flow between the adjacent zones is achieved via a sealing member **900** that comprises a metallic spring assembly having first and second undulating side sections **912** and **916** that are arranged symmetrical to one another, as shown in FIG. **9**. End portions **912***a* and **916***a* of the undulating side sections **912** and **916** are attached or coupled to the radial plate **37**. A second end portion or extension **920** of the sealing member **900** is disposed between the undulating side sections **912** and **916**. The undulating side sections **912** and **916** are configured to bias the second end portion **920** toward the sector plate **29** to press against the first surface **29***a* of the sector plate **29** when the second surface **37***a* of the radial plate **37** is aligned with the first surface **29***a* of the sector plate **29**, thereby minimizing or preventing fluid flow between the adjacent zones. According to some implementations, the second end portion **920** includes leading and trailing ramp surfaces **922** and **924** that facilitate a smooth compression and decompression of the sealing member **900** as the sealing member **900** respectively approaches and moves away from the sector plate **29**. When the second surface **37***a* of the radial plate **37** is aligned with the first surface **29***a* of the sector plate **29**, at least a portion of the leading ramp surface **922** is located on the first side **37***b* of the radial plate **37** and at least a portion of the trailing ramp surface **924** is located on the second side **37***c* of the radial plate **37**. The symmetric arrangement of the first and second undulating side sections **912** and **916** advantageously continuously urges the second end portion **920** of the sealing member **900** towards a vertical alignment with the radial plate **37**.

[0074] As shown in FIG. **9**, according to some implementations, the second end portion **920** of the sealing member **900** includes first and second of grooves **922***a* and **922***b* that are formed between walls **902** of the second end portion **920**. Each of grooves **922***a* and **922***b* resides in an area between the leading and trailing ramp surfaces **922** and **924** and has an opening **904** that faces the first surface **29***a* of the sector plate **29**. In the example of FIG. **9**, the first and second grooves **922***a* and **922***b* are separated by a central wall **925** that is arranged orthogonal to the first surface **29***a* of the sector plate **29**. In other implementations, the central wall may be arranged non-orthogonal to the first surface **29***a*. The purpose of grooves **922***a* and **922***b* is create fluid flow expansions, fluid flow contractions, and/or fluid flow vortices in the event that fluid flow is inadvertently established across the intended contact region of the second end portion **920** of the sealing member **900** with the first surface **29***a* of the sector plate **29**. According to other implementations, the second end portion **920** of the sealing member **900** is devoid of grooves and may possess a flat or substantially flat surface that is configured to press against the first surface **29***a* of the sector plate **29**.

[0075] According to some implementations, each of the first and second undulating side sections **912** and **916** has first protruding members or extensions **913** and **917** that have vertically spaced-apart surfaces **913***a*/**913***b* and **917***a*/**917***b* that each face towards the first surface **29***a* of the sector plate **29** when the second end portion **920** of the metallic spring assembly is pressed against the first surface **29***a* of the sector plate **29**. According to some implementations, the first and second undulating side sections **912** and **916** additionally include second protruding members or extensions **914** and **918** that are respectively vertically spaced-apart from and located between the radial plate **37** and the first protruding members **913** and **917**. The second protruding members **914** and **918** have vertically spaced-apart surfaces **914***a*/**914***b* and **918***a*/**918***b* that each face towards the first surface **29***a* of the sector plate **29** when the second end portion **920** of the sealing member **900** is pressed against the first surface **29***a* of the sector plate **29**.

[0076] According to some implementations, surfaces **913***a-b*, **914***a-b*, **917***a-b*, and **918***a-b* of the protruding members **913**, **914**, **917**, and **918** are each arranged parallel or substantially parallel (within ±15 degrees of parallel) to the first surface **29***a* of the sector plate **29** when the second surface **37***a* of the radial plate **37** is aligned with the first surface **29***a* of the sector plate **29**. According to some implementations, the sealing member **900** is a unitary structure being made from a single piece of metal.

[0077] According to other implementations, the minimizing or prevention of fluid flow between the adjacent zones is achieved through the use of a sealing member **950** that includes a metallic spring assembly **960** and a non-spring element **980** like that shown in FIG. **10**. According to some implementations, the metallic spring assembly **960** is a unitary structure being made from a single piece of metal and the non-spring element **980** is also a unitary structure that is made from another single piece of material (e.g., a polymeric material).

[0078] The metallic spring assembly **960** includes first and second undulating side sections **962** and **966** that are arranged symmetrical to one another and that are respectively similar in shape and function to the first and second undulating side sections **912** and **916** described above in conjunction with the example of FIG. **9**. End portions **962***a* and **966***a* of the undulating side sections **962** and **966** are attached or coupled to the radial plate **37**. As shown in FIG. **10**, a proximal end portion of the non-spring element **980** is coupled to and supported by the metallic spring assembly **960**. The metallic spring assembly **960** biases the non-spring element **980** toward the sector plate **29** to press a distal end portion or extension **990** of the non-spring element **980** against the first surface **29***a* of the sector plate **29** when the second surface **37***a* of the radial plate **37** is aligned with the first surface **29***a* of the sector plate **29**. The non-spring element **980** includes leading and trailing ramp surfaces **982** and **984** that facilitate a smooth compression and decompression of the metallic spring assembly **960** as the sealing member **950** respectively approaches and moves away from the sector plate **29**. When the first surface **29***a* of the radial plate **29** is aligned with the second surface **37***a* of the sector plate **37**, at least a portion of the leading ramp surface **982** is located on the first side **37***b* of the radial plate **37** and at least a portion of the trailing ramp surface **984** is located on the second side **37***c* of the radial plate **37**.

[0079] According to some implementations, the non-spring element **980** is made of a rigid material that does not deform when the surface of the distal end portion **990** is pressed against the first surface **29***a* of the sector plate **29**. According to some implementations, the non-spring element **980** is a solid structure, whereas in other implementations, the non-spring element **980** is a hollow structure. The non-spring element **980** may be made of a material that is more wear resistant than the metallic spring assembly **960** and can be readily replaceable. Regarding the latter, as shown in the example of FIG. **10**, attachment of the non-spring element **980** to the metallic spring assembly **960** can occur by sliding a non-circular shaped distal end protuberance **970***a* of the metallic spring assembly **960** into a mating slot **970***b* inside the non-spring element **980**. In the example of FIG. **10**, the non-spring element **980** is keyed to the metallic spring assembly **960** in a way that prevents or otherwise limits rotation of the non-spring element **980** with respect to the metallic spring assembly **960** to no more than ±10 degrees. The implementation of the non-spring element **980** may provide adequate sealing between the radial plate **37** and the sector plate **29** while increasing a useful lifespan of the sealing member **950**.

[0080] According to some implementation, the distal end portion **990** of the non-spring element **980** includes first and second grooves **991***a* and **991***b* that are formed like and function like the grooves **922***a* and **922***b* described above in conjunction with the example of FIG. **9** (e.g., to create fluid flow expansions, fluid flow contractions, and/or fluid flow vortices in the event that fluid flow is inadvertently established across the intended contact region of the non-spring element **980** with the first surface **29***a* of the sector plate **29**). However, according to other implementations, the distal end portion **990** is devoid of grooves and possesses a flat or substantially flat surface that is configured to press against the first surface **29***a* of the sector plate **29**.

[0081] Each of the sealing members **501**, **900**, **950** and flow restrictors **600**, **700** can be considered a flow restrictor or inhibitor that prevents or at least discourages fluid flow between the flow restrictor and a part (e.g., a plate) of a housing. In particular, the sealing members **501**, **900**, **950** are configured to abut the housing to inhibit fluid flow across the rotor, whereas the flow restrictors **600**, **700** are configured to disrupt fluid flow across the housing to inhibit fluid flow across the housing.

[0082] Additionally, any of the sealing members **501**, **900**, **950** and/or flow restrictors **600**, **700** can be attached to a rotor (e.g., the rotor **34**) in various manners to block undesirable fluid flow across or around the rotor, such as between adjacent zones. FIG. **11** illustrates one example by illustrating a portion of the RAM **26** in greater detail. The housing **100** of the RAM **26** includes an inner structure **1000** about which the rotor **34** rotates. For example, the inner structure **1000** may include a hub. The radial plates **37** of the rotor **34** extend radially outward from the inner structure **1000**. A flow restrictor **1002** (e.g., any of the sealing members **501**, **900**, **950** and/or flow restrictors **600**, **700**) may be attached to a proximal end **1003** of at least one of the radial plates **37** to inhibit fluid flow between the radial plates **37** and the inner structure **1000**. For example, the flow restrictor **1002** may contact (e.g., sealingly engage with) the inner structure **1000** and/or disrupt fluid flow between the rotor **34** and the inner structure **1000**.

[0083] The housing **100** of the RAM **26** further includes an outer wall **1004** surrounding (e.g., circumferentially surrounding) a perimeter of the rotor **34**. For instance, at least a portion of the outer wall **1004** may be positioned radially outward of the rotor **34** and/or may interconnect top and bottom sector plates of the RAM **26**. In the depicted embodiment, the outer wall **1004** includes an outer plate **1006** extending in overlap with a portion of the rotor **34**, such as a distal end of the radial plates **37**. A flow restrictor **1002** may additionally or alternatively be coupled to the rotor **34** to inhibit fluid flow between the outer wall **1004** and the rotor **34**.

[0084] As an example, a flow restrictor **1002** may be coupled to a distal end **1008** of at least one of the radial plates **37** of the rotor **34** and extend (e.g., axially extend) between the rotor **34** and the outer plate **1006**. The flow restrictor **1002** may contact (e.g., sealingly engage with) the outer plate **1006** and/or disrupt fluid flow between the radial plates **37** and the outer plate **1006** to inhibit fluid flow between the radial plates **37** and the outer plate **1006**. This may prevent or inhibit leakage between adjacent zones of the RAM **26** while also preventing or inhibiting circumferential leakage around a side of the rotor **34**.

[0085] Furthermore, the rotor **34** includes distal plates **1010** attached to the distal end **1008** of the radial plates **37** and extending between adjacent radial plates **37**. A flow restrictor **1002** may be coupled to one of the distal plates **1010** and extend (e.g., axially extend) between the rotor **34** and the outer plate **1006** to inhibit fluid flow between the rotor **34** and the outer plate **1006**. Additionally or alternatively, the flow restrictor **1002** may be coupled to a dedicated component (e.g., a seal carrier bar) that is positioned outward of the distal plates **1010** to extend (e.g., axially extend) between the rotor **34** and the outer plate **1006** and inhibit fluid flow between the rotor **34** and the outer plate **1006**. For example, in either case, the flow restrictor **1002** may contact (e.g., sealingly engage with) the outer plate **1006** and/or disrupt fluid flow between the distal plates **1010** and the outer plate **1006** to inhibit fluid flow between the distal plates **1010** and the outer plate **1006**. Again, this may prevent or inhibit leakage between adjacent zones of the RAM **26** while also preventing or inhibiting circumferential leakage around a side of the rotor **34**.

[0086] FIG. **12** illustrates another example of the implementation of the flow restrictor **1002** by illustrating another portion of the RAM **26** in greater detail. The housing **100** of the RAM **26** includes the outer wall **1004**, which includes an axial plate **1020** positioned radially outward of the rotor **34**, such as beyond a circumference of the rotor **34**, between top and bottom sector plates, and/or extending along at least a portion of the outer shell **35** (e.g., a circumference of the outer shell **35**) of the rotor **34**. In the illustrated embodiment, the radial plates **37** of the rotor **34** and/or radially extending flanges extend radially beyond the outer shell **35** such that the distal end **1008** of a radial plate **37** or of a radially extending flange extends (e.g., radially extends) between the outer shell **35** and the axial plate **1020** when the radial plate **37** is aligned with the axial plate **1020**. A flow restrictor **1002** may be attached to the distal end **1008** and contact (e.g., sealingly engage with) the outer plate **1006** and/or disrupt fluid flow between the radial plates **37** and the axial plate **1020** to inhibit fluid flow between the radial plates **37** and the axial plate **1020**. Alternatively, flow restrictor(s) **1002** may be installed in similar locations even if the radial plates **37** do not protrude/extend as shown.

[0087] Overall, the RAM implementations provided herein achieve at least the advantages described herein. However, to be clear, while the application utilizes specific implementations to describe the RAM, as well as the advantages thereof, it is not intended to be limited to the details shown. Instead, it will be apparent that various modifications and structural changes may be made therein without departing from the scope of the inventions and within the scope and range of equivalents of the claims. In addition, various features from one of the implementations may be incorporated into another of the implementations.

[0088] It is also to be understood that the sector plate described herein, or portions thereof may be fabricated from any suitable material or combination of materials, such as metals or synthetic materials including, but not limited to, plastic, rubber, derivatives thereof, and combinations thereof. It is also intended that the present invention cover modifications and variations of this invention. For example, it is to be understood that terms such as “left”, “right”, “top”, “bottom”, “upper”, “lower”, “front”, “rear”, “side”, “height”, “length”, “width” “interior”, “exterior,” “inner”, “outer” and the like as may be used herein, merely describe points of reference and do not limit the present invention to any particular orientation or configuration.

[0089] Finally, when used herein, the term “comprises” and its derivations (such as “comprising”, etc.) should not be understood in an excluding sense, that is, these terms should not be interpreted as excluding the possibility that what is described and defined may include further elements, steps, etc. Meanwhile, when used herein, the term “approximately” and terms of its family (such as “approximate”, etc.) should be understood as indicating values very near to those which accompany the aforementioned term. That is to say, a deviation within reasonable limits from an exact value should be accepted, because a skilled person in the art will understand that such a deviation from the values indicated is inevitable due to measurement inaccuracies, etc. The same applies to the terms “about” and “around” and “substantially”.

Claims

1. A rotary machine, comprising: a rotor with a plurality of plates defining openings therebetween; a housing enclosing the rotor; and a flow restrictor comprising: a body attached to a plate of the plurality of plates of the rotor and extending from the rotor toward the housing; and a plurality of extensions extending from the body toward the housing to form a groove between extensions of the plurality of extensions.

2. The rotary machine of claim 1, wherein an extension of the plurality of extensions is configured to flex toward the housing in response to a pressure differential across the rotor to provide a seal against the housing.

3. The rotary machine of claim 1, wherein an extension of the plurality of extensions comprises: an angled surface extending obliquely from the body toward the housing; and a distal surface that is offset from the housing to form a gap between the housing and the extension, wherein the angled surface is configured to direct fluid flow toward the distal surface to deflect fluid flow advancing toward the gap.

4. The rotary machine of claim 1, wherein the plurality of extensions comprises a first extension extending obliquely from the body toward the housing and a second extension extending orthogonally toward the housing to form the groove between the first extension and the second extension, and the groove is configured to form fluid flow vortices to disrupt fluid flowing between the housing and the flow restrictor.

5. The rotary machine of claim 1, wherein the rotary machine comprises a plurality of zones, each zone of the plurality of zones is configured to receive a different fluid flow, the housing comprises a sector plate extending along a surface of the rotor between adjacent zones of the plurality of zones, and the flow restrictor extends between the plate of the plurality of plates of the rotor and the sector plate when the plate is in alignment with the sector plate.

6. The rotary machine of claim 1, wherein the rotary machine comprises an outer wall positioned radially beyond the rotor, and the flow restrictor extends between the plate of the plurality of plates of the rotor and the outer wall.

7. The rotary machine of claim 1, wherein the housing comprises an inner structure about which the rotor is configured to rotate, and the flow restrictor extends between the plate of the plurality of plates of the rotor and the inner structure.
8. The rotary machine of claim 1, wherein the body comprises a spring configured to bias the plurality of extensions toward the housing.
9. The rotary machine of claim 1, comprising adsorbent material disposed in the openings defined between the plurality of plates.
10. The rotary machine of claim 1, wherein the rotary machine comprises a rotary heat exchanger.
11. A flow restrictor for a rotary machine, the flow restrictor comprising: a body configured to attach to a rotor of the rotary machine, the rotor comprising a plurality of plates, and the body being configured to attach to a plate of the plurality of plates; and a plurality of extensions extending from the body toward a housing of the rotary machine to inhibit fluid flow between the housing and the plate.
12. The flow restrictor of claim 11, wherein an extension of the plurality of extensions is curved to form a convex surface facing the housing and a concave surface opposite the convex surface.
13. The flow restrictor of claim 11, wherein an extension of the plurality of extensions extends obliquely outward from the body toward the housing of the rotary machine.
14. The flow restrictor of claim 11, wherein the body comprises a spring configured to bias the plurality of extensions toward the housing.
15. The flow restrictor of claim 14, wherein the body and the plurality of extensions are separate components coupled to one another.
16. The flow restrictor of claim 15, wherein the body and the plurality of extensions are composed of different materials.
17. A flow restrictor of a rotary machine, the flow restrictor comprising: a body configured to attach to opposite sides of a plate of a rotor of the rotary machine; and an extension extending from a distal end of the body, wherein the extension is configured to flex toward a housing of the rotary machine in response to a pressure differential across the rotor to provide a seal against the housing.
18. The flow restrictor of claim 17, wherein the extension extends from the body in a direction of rotation of the rotor.
19. The flow restrictor of claim 18, comprising an additional extension extending from the distal end of the body in an opposite direction of rotation of the rotor, wherein the extension and the additional extension form a groove therebetween.
20. The flow restrictor of claim 17, wherein the extension is curved to form a concave surface facing the body and a convex surface facing away from the body.